Activity 1 : Gray-level Transformations

① Medical Image Enhancement.

1. Which gray-level transformation would you recommend?

Contrast streching trans

2. Why is it suitable?

It improves the contrast of darker tissues and makes hidden details more visible.

3. What effect will it have on the stage?

The tissues become clearer, and the difference between bones and surrounding tissues are easier to identify.

② Night Time Serveillance :-

1. Which transformation would improve the visibility?

Logarithmic transformation.

2. Why is it more suitable than negative transformation?

log transformation enhances dark pixel values, whereas negative transformati

only reverses brightness levels.

3. Predict the appearence of the enhanced image.

The image becomes brighter, making vechiles, people, and other objects more visible.

(3) Fingerprint Recognition :-

1) Which gray-level transformation would you apply?

Contrast stretching transformation.

2) Explain why.

It increases the contrast between fingerprint ridges and valleys, improving fingerprint recognition

3) What would happen if a log transformation were applied instead?

It brightens dark regions and compresses bright values, but may not improve

ridge-valley contrast as effectively as contrast streching.